

Unit 10, Lesson 4: Solving Problems Involving Arc Length and Area of a Sector



Benchmark

MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.



Learning Target

- ☐ I can solve mathematical and real-world problems involving arc length and sector area.



10.4.1 Warm-Up: Bell Work

1. A small, family-owned restaurant sells a personal pan pizza with a radius of 10 centimeters for \$5. A nearby restaurant cuts their pizza with a radius of 30 centimeters into 8 slices of equal size, and charges \$5 per slice.

Which is a better deal? Explain your reasoning using mathematics.



10.4.2 Exploration: Finding a Formula for the Area of a Sector

1. Suppose a circle is cut into 8 equal slices to form sectors.

a. Complete the table.

Number of Slices	Fraction of the Circle	Central Angle in Degrees ($^{\circ}$)	Write an Equation to Find the Area of the Sector
2			
4			
6			
8			

- b. Compare the values in your table with your partner. Discuss your equations in the last column and make any revisions if needed.

- c. Discuss with your partner how the fractions in your equations relate to the central angles, and how the equations might have differed if the angle measure was given in radians.

- d. Summarize your discussion by writing a general formula to find the area of any sector.



Area of a Sector

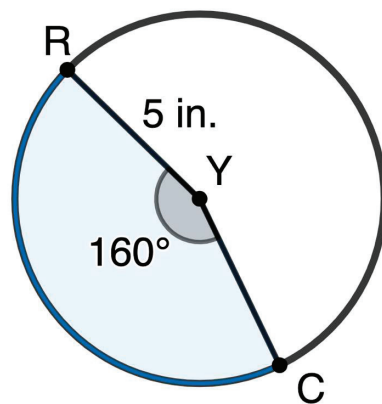
$$A = \frac{\theta}{360} \pi r^2, \text{ where } \theta \text{ is in degrees}$$

$$A = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$



10.4.3 Guided Practice: Area of a Sector

1. Circle Y has radius $\overline{CY}$ that measures 5 inches with central angle $\angle RYC$, where $m\angle RYC = 160^\circ$. Find the area of the shaded sector. Round your answer to the nearest thousandth.



2. Circle X has a diameter of 12 centimeters. Points B and P are on the circle and form the sector BXP . The central angle $\angle BXP$ in Circle X measures 60° . Determine the area of the sector. Express your answer in terms of π .
3. The area of a sector with a radius of 8 inches is 74.84 square inches. Calculate the approximate measure of the central angle of the sector in degrees.



10.4.4 Collaboration: Arc Length

Work with your partner to complete the following.

1. Explain how the circumference relates to an arc length of a circle.
2. Describe how to find the arc length of half a circle.
3. Describe how to find the arc length of a quarter of a circle.
4. Describe how to find the arc length of a circle with a central angle of 60° .

5. Write a general equation that can be used to find the arc length of a sector in degrees.

6. Discuss with your partner how your equation would change if the central angle had been given in radians. Summarize your discussion by writing an equation to find the arc length if the angle measure is given in radians.



Arc Length

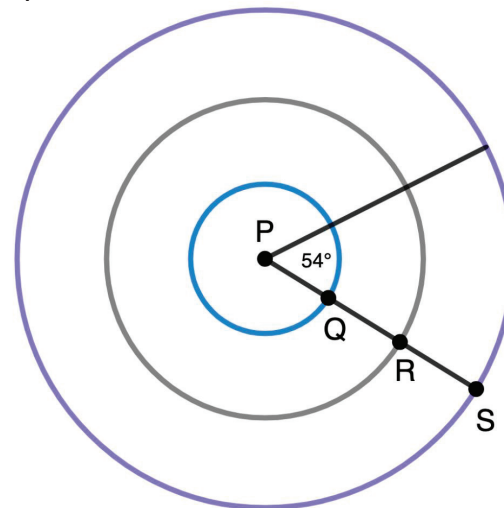
Arc Length = $C \cdot \frac{\theta}{360^\circ}$, where θ is in degrees

Arc Length = $r \cdot \theta$, where θ is in radians



10.4.5 Guided Practice: Arc Length

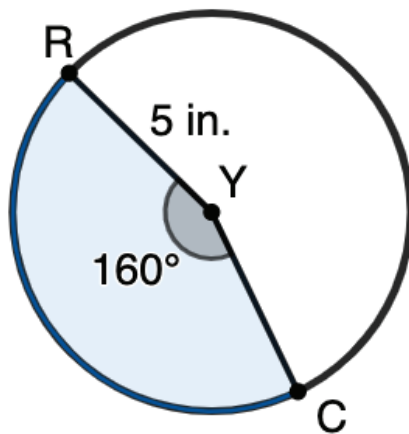
1. Concentric circles with center P are shown with $m\angle P = 54^\circ$. The smallest Circle P has a radius $\overline{PQ}$ that measures 15 meters (m). The largest Circle P has radius $\overline{PS}$ and the middle Circle P has radius $\overline{PR}$. Each circle has a radius that is 15 m longer than the immediate prior circle.



Calculate the arc length intercepted by $\angle P$ on each circle.

Arc length of circle with radius $\overline{PQ}$	
Arc length of circle with radius $\overline{PR}$	
Arc length of circle with radius $\overline{PS}$	

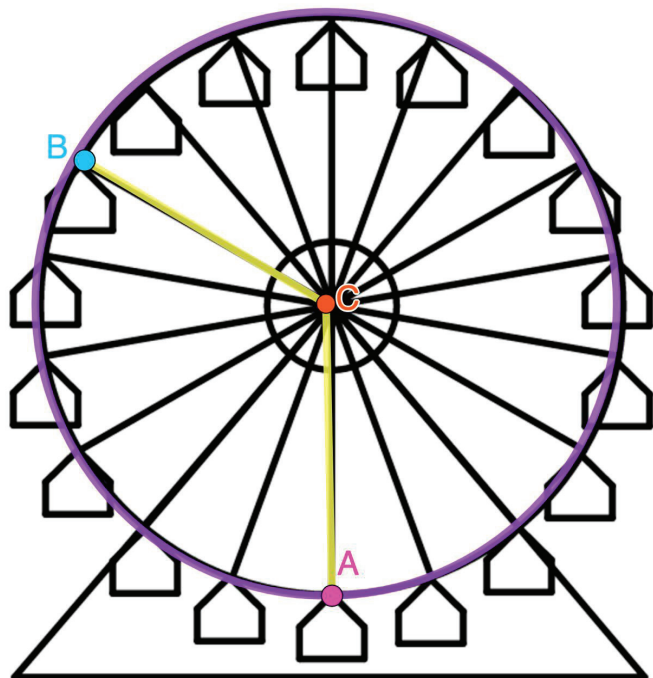
2. Points R and C are on Circle Y . If $m\angle CYR$ is 160° , find the length of $\widehat{RC}$.



10.4.6 Guided Practice: Solving Real-World Problems Involving Arc Length and Area of a Sector

1. In Sarah's family, the person celebrating their birthday always gets to cut the first piece of a circular birthday cake. Sarah is celebrating her birthday with both of her parents, her two brothers and her best friend. She cuts her piece of birthday cake and it formed a sector with a diameter of 8 inches and a central angle with measure 85° . If the 5 other party guests equally share the remaining portion of the cake, what is the approximate area of cake each person receives? Round your answer to two decimal places.

2. Dominic is riding a Ferris wheel at the county fair. The Ferris wheel forms a circle with center C . The Ferris wheel has a radius of 60 ft and rotates clockwise with seats that are equidistant from each other.



- a. After the ride begins from point A and travels **clockwise**, it pauses at point B . How many feet has Dominic traveled when his seat pauses at point B ? Write your answer in π form and rounded to the nearest thousandth.

π or feet

- b. About how many feet has Dominic traveled if the full ride is 4 revolutions?

π or feet



10.4.7 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.